

LEVEL 1

In this level, your challenge is to build a **working Web based car park management system** that detects arriving cars, guides them to the right parking spots, handles payments at exit, and keeps the basic car park operation under control.

Start by reading the documentation and exploring the **simulator** to understand how vehicles, parking spots, gates, and events work. Then plan your **dashboard and web interface**, deciding what operators need to monitor and control.

Next, set up your **web server, webhook listener, and database**. Connect your system to the simulator by calling its APIs, receive webhook events, and send the appropriate commands.

Your Webserver is used to show dashboard and other pages, make api calls ... ,
Webhook is to receive server calls and store in database for further process/dispay ..

Your system should be able to:

- Call the simulator API to login, then to inspect and control the car park, such as **listing parking spots** and **opening/closing gates**.
- Receive and handle simulator events, such as **a car arriving at the entrance**.
- Allow cars to enter and guide them to available parking spots.
- Handle car exits and calculate the correct parking charges (refer to documentaions). then Ask car to leavepark
- Create Website with dashboard to show the status of the system.
- Keep parking data and the dashboard up to date. Like state of gates and parking spots.
- Display **occupied/free parking spots by zone** and the current **gate status**. So new cars can leave if park has no free spots.
- Provide basic authentication and roles such as **Admin** and **Operator**. Operator can manually control the park components and see the status of gates and spots
- Log car arrivals/parking time/departure/charges in database, so dashboard can show and search these events. effeciently
- Don't forget the presentation for last judging stage

The Goal

Keep the cars moving, parking spaces organized, and the car park running smoothly.
You win when park work as expected with its full capacity.